

Tahira Kazimi

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Introduction

Ph.D. student in Computer Science at Virginia Tech specializing in Generative AI and Computer Vision, with a focus on advancing alignment, personalization and video/action understanding in generative systems. First-author at CVPR 2025 on interpretability of diffusion models and CVPR 2026 on diverse video generation; recognized as a *Rising Star* at MIT with background spanning reasoning agents, physically-grounded, diverse, and video to video generation tasks. Prior experience with Gaussian Splatting, NeRF and diffusion-based 3D/4D rendering.

Education

Virginia Tech (VA), Ph.D. in Computer Science August 2024 – May 2028

- GPA: 4.0/4.0
- Advisor: Prof. Pinar Yanardag (GEMLab website)
- **Coursework:** Generative Modeling, Interpretability, Representation Learning

Middle East Technical University (METU) (TURKEY), BSc in Computer Engineering Sept 2019 – June 2024

- GPA: 3.5/4.0 (transcript)
- **Coursework:** Deep Learning, Artificial Intelligence, Data Structures, Algorithms, Operating Systems

Publications

HiPhy: Hierarchical Alignment for Physically-Plausible Multi-Principle Video Generation NeurIPS 2026

Tahira Kazimi, Shubhankar Borse, Munawar Hayat, Fatih Porikli, Pinar Yanardag

NeurIPS 2026

Project page

Diverse Video Generation with Determinantal Point Process-Guided Policy Optimization CVPR 2026

Tahira Kazimi, Connor Dunlop, Pinar Yanardag

CVPR 2026

Project page

Explaining in Diffusion: Explaining a Classifier Through Hierarchical Semantics with Text-to-Image Diffusion Models CVPR 2025

Tahira Kazimi, Ritika Allada, Pinar Yanardag

CVPR 2025

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Diverse Video Generation with Collaborative Agents IEEE CAI 2026, ICCV 2025 workshop

Tahira Kazimi, Heather Yu, Zhiqiang Lao

ICCV 2025 workshop

Audit & Repair: An Agentic Framework for Consistent Story Visualization in Text-to-Image Diffusion Models ICCV 2025 workshop

Kiyomet Akdemir*, Tahira Kazimi*, Pinar Yanardag

ICCV 2025 workshop

Project page

Experience

Machine Learning Engineering Intern , Apple Inc. – San Diego, CA	May 2026 – August 2026
• Worked on diffusion based video to video tasks. Trained diffusion based video inpainting, outpainting, and super-resolution models.	
Generative AI and Computer Vision Intern , Futurewei Technologies – Basking Ridge, NJ	May 2025 – August 2025
• Developed a reasoning planning agentic system that deploys models as tools to enhance the visual diversity in generated outputs. • Demonstrated that the proposed pipeline outperforms existing baselines in both diversity and visual quality by 15–20%, enabling richer creative outputs for content generation workflows, while preserving semantic consistency (ICCV’25 workshop).	
Research and Development Engineer , IKON ARGE Teknoloji – Ankara, TURKEY	Jan 2024 – Aug 2024
• Engineered an LLM-powered virtual assistant system to manage visitor interactions at office entrances, automating the process of routing visitor requests to appropriate departments and notifying the correct employee. • Developed reinforcement learning agents capable of detecting goal positions from a signal sent from goal position, utilizing Proximal Policy Optimization (PPO). • Trained and implemented an Automatic Plate Detection model on NVIDIA JETSON hardware to enable real-time processing and object recognition using YOLOv8.	
Undergraduate Research Assistant , Image Processing and Pattern Recognition Lab – Ankara, TURKEY	Sept 2023 – May 2024
• Principal investigator: Prof. Emre Akbas (personal website) • Conducted research on texture bias in ImageNet-trained models, developing shape-based object recognition models that utilized 3D shape representations for recognition tasks using NeRF and Gaussian Splatting. • Developed a segmentation model for medical image analysis improving segmentation accuracy and adaptability across diverse datasets.	
Research Intern , TUBITAK – Ankara, TURKEY	July 2023 – Sept 2023
• Principal investigator: Prof. Sinan Kalkan (personal website) • Conducted photovoltaic energy estimation from solar panels using time-series deep learning models, enhancing forecasting accuracy and supporting more reliable energy output predictions for renewable energy systems. • Trained Long Short-Term Memory (LSTM) and Neural Hierarchical Interpolation for Time Series (N-HiTs) models on photovoltaic energy datasets.	

Academic Activities and Awards

Intern Award, IC lab : Received the <i>Best Intern Award</i> in recognition of outstanding research impact and technical contributions.	Dec 2025
Rising Star MIT : Selected to present latest research at MIT as part of the Rising Star program, with an acceptance rate of 20% . Selected awardee page	Sept 2025
Reviewer AAAI, CVPR, CVPR Responsible GenAI Workshop Workshop Page	2024 - Present
P13N Workshop Organizer and Reviewer , ICCV Personalization in Generative AI Workshop Workshop Page	June 2025

Projects

SilhouetteNet: Shape-based object recognition	poster link
• Research project developed as part of the guided research course at METU • Tackled texture bias in ImageNet-trained CNN models by developing classifiers capable of recognizing objects	

based solely on their physical shape, independent of texture information. This new pipeline was based on an R-CNN instance segmentation architecture.

- Enhanced model performance by 15% through fine-tuning ImageNet-trained models using image-level labels in a semi-supervised learning framework.
- Demonstrated that ImageNet-pretrained models possess the capacity to learn shape representations when explicitly provided with shape-based input.

Planify: AI scheduler

project website

- Collaborated with a team of five to develop a personalized AI-driven scheduler application aimed at effectively organizing users' schedules based on preferences learned from their app interactions.
- Collaborated in developing the objective function for optimizing and scheduling the tasks for one week using constraint programming.
- Designed the chatbot for adding events into calendar using text and voice commands with text extraction models.
- Assisted with front-end development in React Native using Node.js, creating the app's functionality in JavaScript, employed HTML for webpage structure and CSS for formatting and styling, ensuring a clean, user-friendly interface.
- Tools Used: Google's OR-Tools, Flask, React

Technologies

Languages: C++, C, Java, Python, SQL, PyTorch, CUDA, Docker, Git, Object Detection, Semantic/instance segmentation

Technologies: Programming & Tools: Python, C++, Java, SQL, CUDA, Docker, Git
ML/AI Frameworks: PyTorch, Diffusion Models, RL, NLP, CV (YOLO, R-CNN, UniverSeg)

Languages: Dari, English, Turkish